

Technische Hochschule Ingolstadt
Fakultät Elektro- und Informationstechnik

Sensor Network Technologies and Sensor Fusion

Multi-Object Tracking

Nov 26, 2025

Welcome to the lecture!

Prof. Dr.-Ing. Björn Giesler

Technische Hochschule Ingolstadt

AIMotion, AI and Sensor Data Fusion

Email: bjoern.giesler@thi.de

- **Please register for the lecture on Moodle!**
  - Link: https://moodle.thi.de/course/view.php?id=10580
  - Key: **SensNT+DF#WS25**
- **News on Moodle:**
  - Slides for the lecture (upcoming lectures marked as WIP and are subject to change!)
  - Formulary, Exercises (Solutions provided after discussion in the lecture)
  - Exercises are optional, one task / sheet per week, but are highly recommended.
    They can be handed in for correction.
  - I will *try* to publish exercises before Wednesday‘s lecture so that initial questions can be discussed on Thursday.
- **Exam: Written exam, 90 minutes**
- **Consultation hours: Please arrange via e-mail to bjoern.giesler@thi.de.**
- **Please also don‘t hesitate to send questions, lecture feedback, change requests, etc.!**

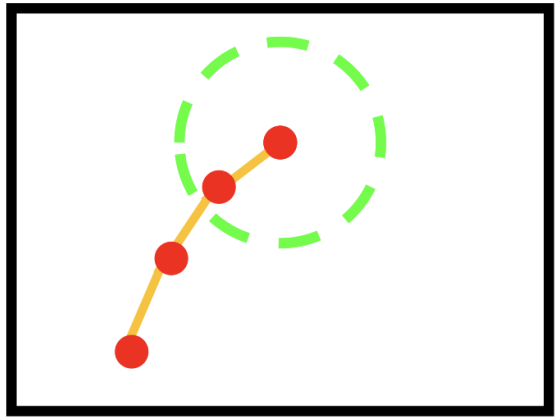
Sensor Networks and Sensor Fusion

Outline

- **In the last lectures we looked at various forms of the Kalman filter – linear, extended, unscented, and explored what to do if we can't *model* a state transition function.**
- **Now we are in the middle of multi-object tracking applications.**
- **Last lecture, we learned about**
  - track assignment using the Mahalanobis distance,
  - the difficulties of creating and maintaining tracks in the presence of noise,
  - and explored the track splitting filter.
- **Going forward, we will check out**
  - Multi Hypothesis tracking (this lecture),
  - tracking by Network Flow Optimization and other global optimization techniques (next lecture)
  - and using DNNs for tracking.
- **We will then learn about Particle Filters and similar inherent multi hypothesis approaches.**
- **After that, we will revisit tracking briefly using Joint Probability Data Association Filters.**
- **The rest of the lecture series will deal with**
  - Simultaneous Localization and Mapping, and
  - (DNN-based) early fusion approaches.

Recap: Track-Splitting Filter

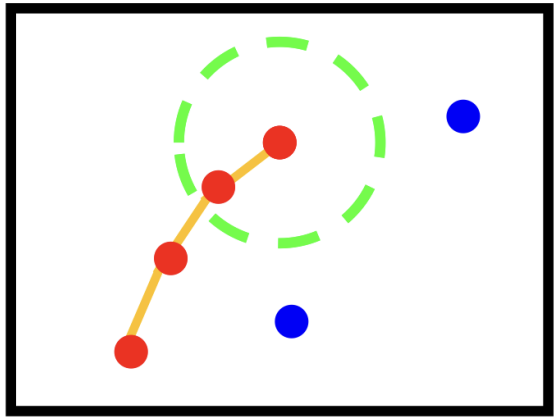
Motion Correspondence



1. Predictions may not be supported by measurements.

   Have the objects ceased to exist?
   ➔ remove them

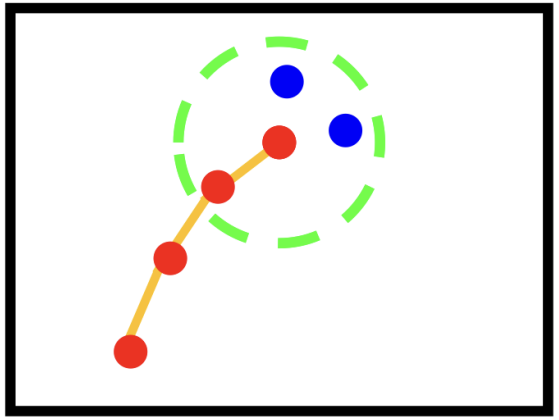
   Are they occluded?
   ➔ keep them



2. There may be unexpected measurements.

   Newly visible objects?
   ➔ set up tracks

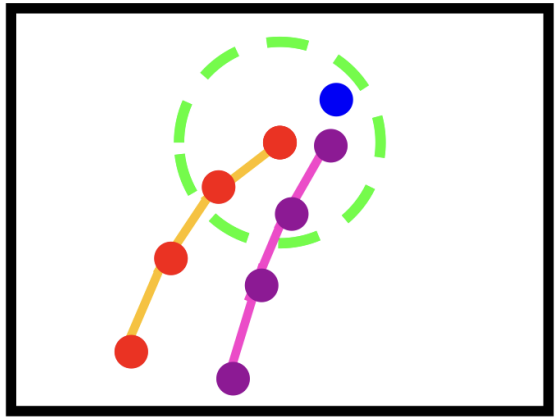
   Just noise?
   ➔ ignore them



3. More than one measurement may match a prediction.

   Which measurement is the correct one?

   What about the others?



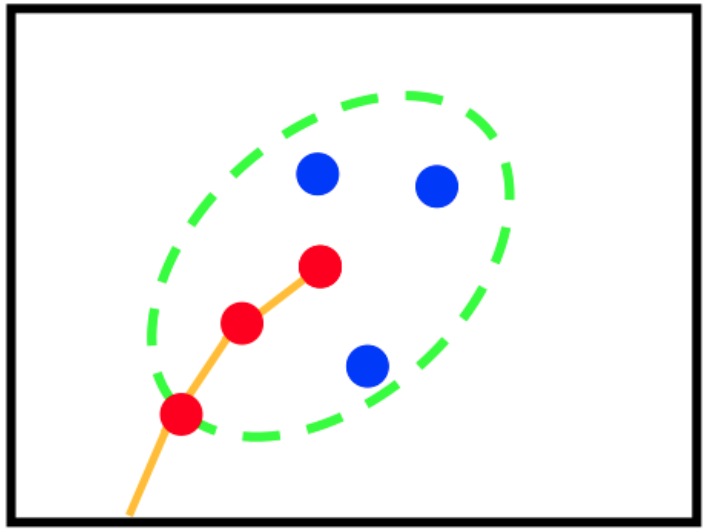
4. A measurement may match to multiple predictions.

   Which track should the measurement be assigned to?

   What about the other track(s)?

Multi-Object Tracking Problem

- We represent a track by a state vector, e.g. $\mathbf{x} = \left[x, y, v_x, v_y\right]^\top$
- As the track evolves, we denote its state by the time index t: $\mathbf{x}^{(t)} = [x^{(t)}, y^{(t)}, v_x^{(t)}, v_y^{(t)}]^\top$
- At each time step, we get a set of observations / measurements: $\mathbf{Y}^{(t)} = \{\mathbf{y}_1^{(t)}, \ldots, \mathbf{y}_{M_t}^{(t)}\}$
- We now need to make the data association between tracks $\{\mathbf{x}_1^{(t)}, \ldots, \mathbf{x}_{N_t}^{(t)}\}$ and measurements $\{\mathbf{y}_1^{(t)}, \ldots, \mathbf{y}_{M_t}^{(t)}\}$:
- Association $z_l^{(t)} = j$ iff $\mathbf{y}_j^{(t)}$ is associated with $\mathbf{x}_l^{(t)}$.



The Mahalanobis Distance as a measure for track similarity

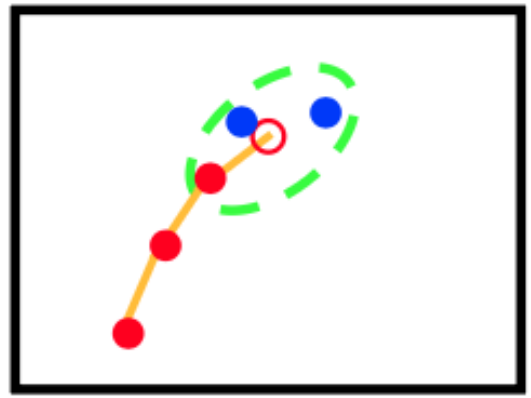
- Our Kalman Filter state of track $\mathbf{x}_l$ is given by the prediction and covariance $\hat{\mathbf{x}}_l^{(t)}, P_l'(t).$
- We have defined the *innovation* that measurement $\mathbf{z}_j$ brings to track $\mathbf{x}_l$ at time t as $\mathbf{i}_{j,l}^{(t)} = (\mathbf{z}_j^{(t)} - \hat{\mathbf{x}}_l^{(t)}).$
- With this, we can write the observation likelihood shortly as $p(\mathbf{z}_j^{(t)}|\mathbf{x}_l^{(t)} \sim \exp\{-\frac{1}{2}\mathbf{i}_{j,l}^{(t)\top} P_l'^{-1}\mathbf{i}_{j,l}^{\top}\}.$

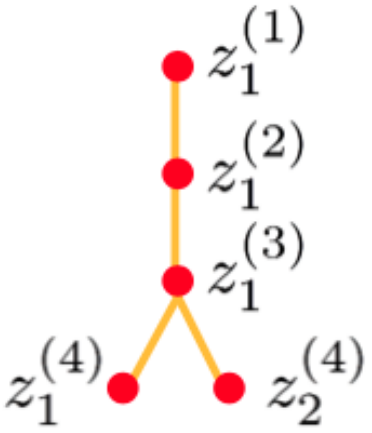
  Mahalanobis Distance
- We can define the ellipsoidal gating or validation volume as $V^{(t)}(\gamma) = \{\mathbf{z} : (\mathbf{y} - \hat{\mathbf{x}}_l^{(t)})^\top P_l'^{(t)-1}(\mathbf{y} - \hat{\mathbf{x}}_l^{(t)}) \leq \gamma\}.$
- This is equivalent to saying "all $\mathbf{y}$ with a Mahalanobis distance $\leq \gamma$ ".

$$z_l^{(t)} = \underset{j}{\operatorname{argmin}}\ (\hat{\mathbf{x}}_l^{(t)} - \mathbf{y}_j^{(t)})^\top P_l'^{-1}(\hat{\mathbf{x}}_l^{(t)} - \mathbf{y}_j^{(t)})$$

Track Splitting Filter

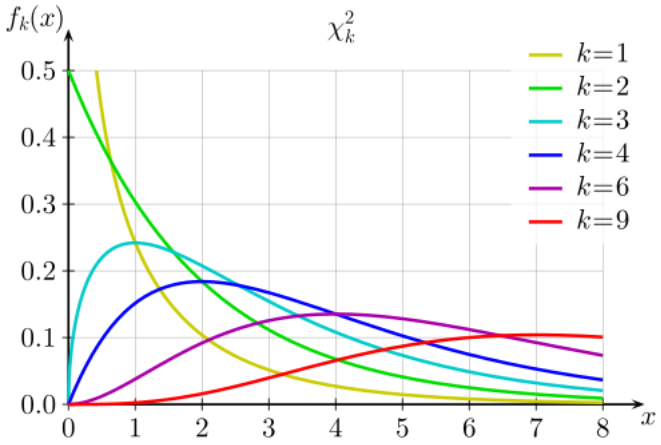
- **The Nearest-Neighbor filter still has uncertainties and options that can't really be resolved**
  - Uncertain decision making
  - ...through gating decision making functions
- **Rather than arbitrarily assigning the "closest" (by whatever norm) measurement, form a tree.**
- **Branches denote alternate assignments. No *decision* is made at this stage; decisions are postponed until additional measurements have been gathered.**
- **Potential problem:**
  Track trees can quickly become very large due to combinatorial explosion.
- **We need some measure of the likelihood so we can prune the tree.**





Track Splitting Filter – Pruning Strategies

- **In order to keep this feasible, need to apply pruning.**
- **Deleting unlikely tracks**
  - May be accomplished by comparing $\lambda(t)$, which follows a χ^2 distribution with $t \cdot n_z$ degrees of freedom, with a threshold α (set according to χ^2 tables).
  - In long tracks, $\lambda(t)$ gets dominated by old terms and responds very slowly to new ones.
  - ➔ Use sliding window to limit track length, or implement exponential decay.
- **Merging track nodes**
  - If the state estimates of two track nodes are similar, merge them.
  - E.g. if both tracks validate identical subsequent measurments.
- **Rank tracks based on $\lambda(t)$, and only keep the most likely N tracks.**



χ^2 distribution for different degrees of freedom

Multi Hypothesis Tracking

Multi Hypothesis Tracking

- **Basic Idea:**
  - Again associate sequences of measurements.
  - Evaluate the probabilities of all association hypotheses.
  - For each sequence of measurements (a hypothesized track), a standard KF yields the state estimate and covariance.
- **Differences to Track-Splitting Filter**
  - Instead of forming a track tree, keep a set of hypotheses that generate child hypotheses based on the associations.
  - After each hypothesis generation step, merge and prune the current hypothesis set to keep the approach feasible.
  - Integrate track generation into the assignment process.
- **Again, a pretty old concept [1] (1979) but still constantly in use today.**

Target vs Measurement Orientation

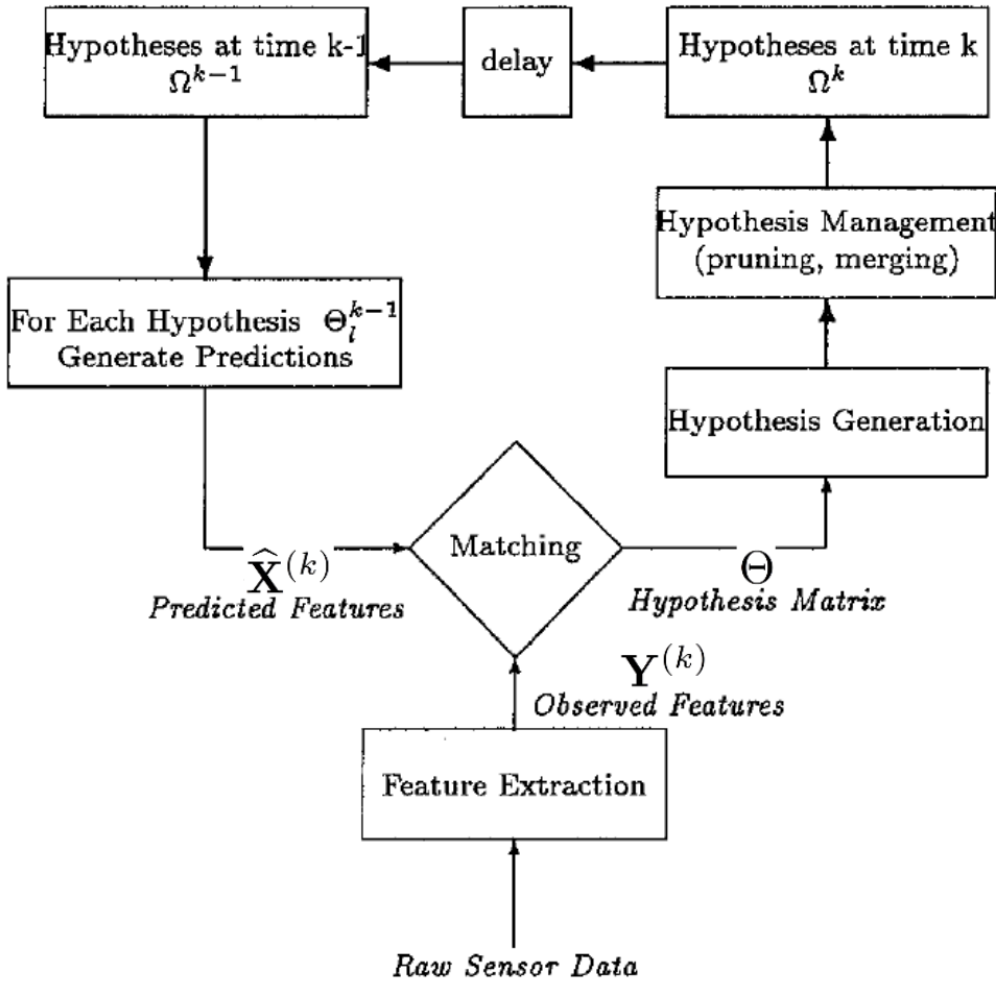
- **Target-oriented approaches**
  - Evaluate the probability that a measurement belongs to an established target.
- **Measurement-oriented approaches**
  - Evaluate the probability that an established target or a new target gave rise to a certain measurement sequence.
  - This makes it possible to include track initiation of new targets within the algorithmic framework.
- **Multi Hypothesis Tracking:**
  - is measurement-oriented
  - Handles track initialization and termination

Strategy

- **Generate all possible hypotheses and then depend on pruning these hypotheses to avoid the combinatorial explosion.**
  - Exhaustive search
  - Tree data structures are used to keep this search efficient
- **Commonly used pruning techniques**
  - Clustering to reduce the combinatorial complexity
  - Pruning of low-probability hypotheses
  - Merging of similar hypotheses
  - N-scan pruning: Select a single best hypothesis at frame K and prune all tracks that do not share the predecessor track at the $(K-N)$th frame.

Ideas

- Instead of forming a track tree,
- keep a set of hypotheses that generate child hypotheses based on the associations.
- Enforce exclusion constraints between tracks and measurements in the assignment.
- Integrate track generation into the assignment process.
- After hypothesis generation, merge and prune the current hypothesis set.



Formalization

- **Set of hypotheses at time t:** $\boldsymbol{\Omega}^{(t)} = \{\Omega_j^{(t)}\}$

- $\boldsymbol{\Omega}^{(t)}$ **is obtained from** $\boldsymbol{\Omega}^{(t-1)}$ **and the latest set of measurements** $\mathbf{Y}^{(t)} = \{\mathbf{y}_1^{(t)}, \ldots, \mathbf{y}_{M_t}^{(t)}\}$

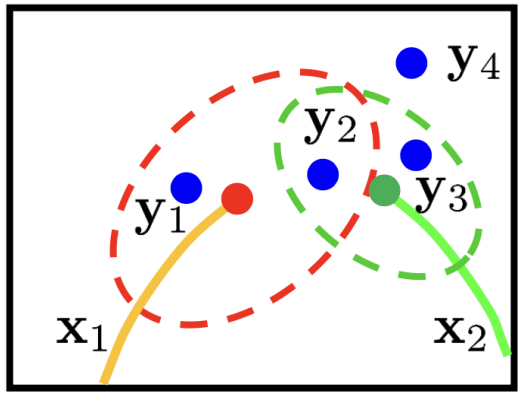
  by performing *all feasible associations* between the old hypotheses and the new measurements.

- **Feasible associations can be**
  - a continuation of a previous track $\mathbf{x}_n$
  - a false alarm $\mathbf{x}_{\text{fa}}$
  - a new target $\mathbf{x}_{\text{nt}}$.

Hypothesis Matrix

- **Visualize feasible associations by a hypothesis matrix**

$$\Theta = \begin{array}{c} \begin{matrix} \mathbf{x}_1 & \mathbf{x}_2 & \mathbf{x}_{\text{fa}} & \mathbf{x}_{\text{nt}} \end{matrix} \\ \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array} \begin{matrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \mathbf{y}_3 \\ \mathbf{y}_4 \end{matrix}$$



- **Interpretation**
  - Columns represent tracked objects
  - Rows represent measurements
  - A non-zero element at (i, j) denotes that a measurement $\boldsymbol{y}_i$ is contained in the validation region of track $\boldsymbol{x}_j$.
  - Extra columns $\boldsymbol{x}_{\text{fa}}$ and $\boldsymbol{x}_{\text{nt}}$ for associations as *false alarm* and *new track*.

Turning feasible associations into assignments

- **For each feasible association, we generate a new hypothesis.**
- **Let**
  - $\Omega_j^{(t)}$ be the j-th hypothesis at time t,
  - $\Omega_{p(j)}^{(t-1)}$ be the parent hypothesis from which $\Omega_j^{(t)}$ was derived, and
  - $Z_j^{(t)}$ denote the set of assignments that gives rise to $\Omega_j^{(t)}$.
- **Assignments are best visualized in matrix form:**

Z_j	$\boldsymbol{x}_1$	$\boldsymbol{x}_2$	$\boldsymbol{x}_{\mathrm{fa}}$	$\boldsymbol{x}_{\mathrm{nt}}$
$\boldsymbol{y}_1$	0	0	1	0
$\boldsymbol{y}_2$	1	0	0	0
$\boldsymbol{y}_3$	0	1	0	0
$\boldsymbol{y}_4$	0	0	0	1

Turning feasible associations into assignments

Z_j	x_1	x_2	x_{fa}	x_{nt}
y_1	0	0	1	0
y_2	1	0	0	0
y_3	0	1	0	0
y_4	0	0	0	1

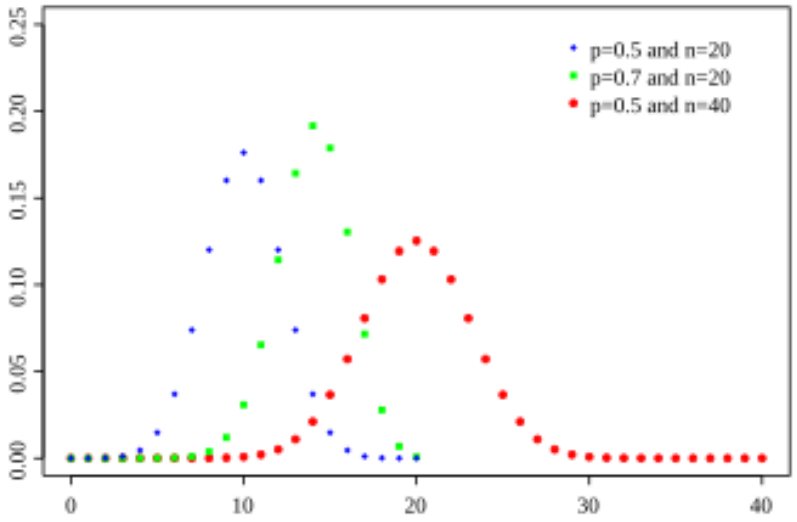
Impose constraints:

- **A measurement can originate from only one object**
  ➔ any *row* has only a single non-zero value.
- **An object can have at most one associated measurement per time step.**
  ➔ any *column* can have only a single non-zero value, *except for* x_{fa} and x_{nt} .

Excursion: Binomial Distribution

- **The Binomial Distribution is the discrete PDF of the number of successes in a sequence of n independent experiments with boolean outcome (success/failure).**
- **Two parameters:**
  - p: Probability of success
  - n: Number of experiments.
- **The probability of getting exactly k successes in n independent trials is given by:**

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

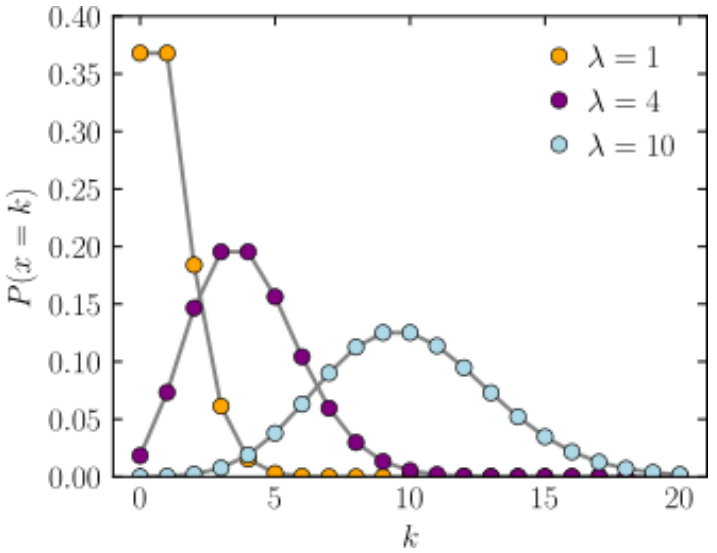


for $k = 0, 1, 2, \ldots, n$ **, with the binomial coefficient** $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

- **Compare to our detection probability:** $p(N_{\text{det}} \mid \Omega_{p(j)}^{(t-1)}) = \binom{N}{N_{\text{det}}} p_{\text{det}}^{N_{\text{det}}} (1 - p_{\text{det}})^{(N - N_{\text{det}})}$

Excursion: Poisson Distribution

- The Poisson Distribution is the discrete PDF expressing the probability of a given number of events occurring in a fixed interval of time, if they occur with a known constant mean rate and independently of the time since the last event.
- One parameter: λ, the expected number of events in a given interval.
- The probability of k events happening in the same given interval is $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$.
- Compare this to our idea of the occurrence of false alarms and new tracks over time!



Probabilistic formulation

- **Enumeration of all possible assignments is easy.**
- **But we also need to calculate the probability of each child hypothesis.**
- **We do this recursively:**

$$
\begin{aligned}
p(\Omega_j^{(t)}|\mathbf{Y}^{(t)}) &= p\left(Z_j^{(t)}, \Omega_{p(j)}^{(t-1)} \mid \mathbf{Y}^{(t)}\right) \\
&= \eta p\left(\mathbf{Y}^{(t)} \mid Z_j^{(t)}, \Omega_{p(j)}^{(t-1)}\right) p\left(Z_j^{(t)}, \Omega_{p(j)}^{(t-1)}\right) \\
&= \eta \underbrace{p\left(\mathbf{Y}^{(t)} \mid Z_j^{(t)}, \Omega_{p(j)}^{(t-1)}\right)}_{\text{Measurement likelihood}} \underbrace{p\left(Z_j^{(t)} \mid \Omega_{p(j)}^{(t-1)}\right)}_{\text{Prob. of assignment set}} \underbrace{p\left(\Omega_{p(j)}^{(t-1)}\right)}_{\text{Prob. of parent}}
\end{aligned}
$$

Normalization factor

Measurement likelihood

Need to determine!

Prob. of assignment set

Need to determine!

Prob. of parent

Know from last cycle!

1. Measurement Likelihood $p\left(\mathbf{Y}^{(t)} \mid Z_j^{(t)}, \Omega_{p(j)}^{(t-1)}\right)$

- **Use Kalman Filter prediction!**
- **Assume that a measurement $\mathbf{y}_i^{(k)}$ associated to a track $\mathbf{x}_j$ is Normal distributed around the measurement prediction $\hat{\mathbf{x}}_j^{(k)}$ with innovation covariance $\hat{\Sigma}_j^{(k)}$.**
- **Assume that the probability distribution of a measurement belonging to a *new track* or *false alarm* is uniform in the observation volume W (the sensor's field of view) with probability W^{-1}.**
- **Then the measurement likelihood can be expressed as**

$$p\left(\mathbf{Y}^{(t)} \mid Z_j^{(t)}, \Omega_{p(j)}^{(t-1)}\right) = \prod_{i=1}^{M_t} \mathcal{N}(\mathbf{y}_i^{(t)}; \hat{\mathbf{x}}_j, \hat{\Sigma}_j^{(t)})^{\delta_i} W^{-(1-\delta_i)}$$

$$= W^{-(N_{\text{fa}}+N_{\text{nt}})} \prod_{i=1}^{M_t} \mathcal{N}(\mathbf{y}_i^{(t)}; \hat{\mathbf{x}}_j, \hat{\Sigma}_j^{(t)})^{\delta_i}.$$

2. Probability of the Assignment Set $p\left(Z_j^{(t)} \mid \Omega_{p(j)}^{(t-1)}\right)$

Composed of three terms.

1. **The probability of the *number of tracks*** $N_{\text{det}}, N_{\text{fa}}, N_{\text{nt}}$ **.**
   1. Assumption: N_{det} follows a ***binomial distribution***:

$$p(N_{\text{det}} \mid \Omega_{p(j)}^{(t-1)}) = \binom{N}{N_{\text{det}}} p_{\text{det}}^{N_{\text{det}}} (1 - p_{\text{det}})^{(N - N_{\text{det}})}$$

      where N is the number of tracks in the parent hypothesis.

   2. Assumption: $N_{\text{fa}}, N_{\text{nt}}$ both follow a ***Poisson distribution*** with expected number of events $\lambda_{\text{fa}} W, \lambda_{\text{nt}} W$:

$$\begin{aligned} &p(N_{\text{det}}, N_{\text{fa}}, N_{\text{nt}} | \Omega_{p(j)}^{(t-1)} \\ &= \binom{N}{N_{\text{det}}} p_{\text{det}}^{N_{\text{det}}} (1 - p_{\text{det}})^{(N - N_{\text{det}})} \cdot \mu(N_{\text{fa}}; \lambda_{\text{fa}} W) \cdot \mu(N_{\text{nt}}; \lambda_{\text{nt}} W) \end{aligned}$$

Probability of the Assignment Set

Composed of three terms.

2. **The probability of a *specific assignment of measurements.***
   - Such that $M_t = N_{\text{det}} + N_{\text{fa}} + N_{\text{nt}}$ holds.
   - This is determined as 1 over the number of combinations

$$\binom{M_t}{N_{\text{det}}} \binom{M_t - N_{\text{det}}}{N_{\text{fa}}} \binom{M_t - N_{\text{det}} - N_{\text{fa}}}{N_{\text{new}}}.$$

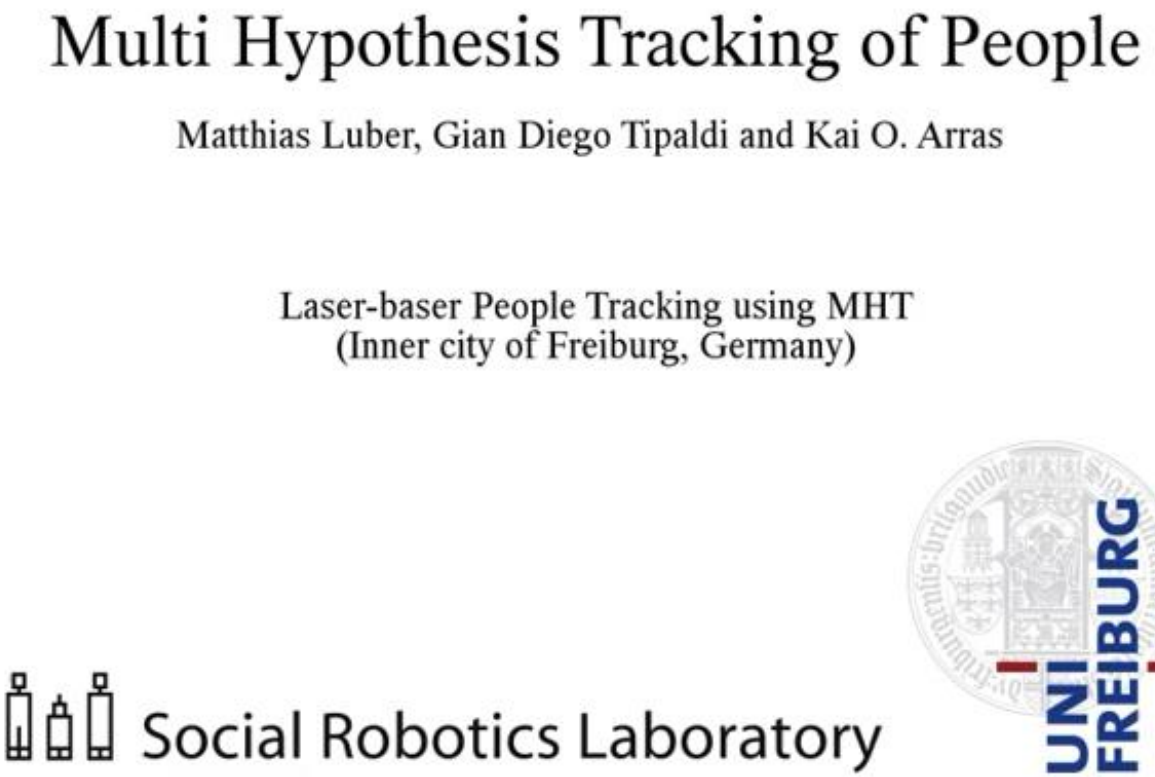
3. **The probability of a *specific assignment of tracks.***
   - Given that a track can be either *detected* or *not detected*.
   - This is determined as 1 over the number of assignments

$$\frac{N!}{(N - N_{\text{det}})!} \binom{N - N_{\text{det}}}{N_{\text{det}}}.$$

Combining all the different parts

- **Nice property: Many terms cancel out! Derivation left as exercise** 😄
- **The final probability** $p(\Omega_j^{(k)}|\mathbf{Y}^{(k)})$ **can be derived in a very simple form.**
- **Practical issues**
  - Exponential complexity remains.
  - Heuristic pruning strategies must be applied to contain the growth of the hypothesis set.
  - E.g. dividing hypotheses into spacially disjoint clusters.

Video: Laser-based People Tracking using MHT



Source:
https://www.youtube.com/watch?v=DQmNyP_oPpQ

Literature / Links

1. **D. Reid: An Algorithm for Tracking Multiple Targets. IEEE Trans. Automatic Control, Vol. 24(6), pp. 843-854, 1979. https://ieeexplore.ieee.org/document/1102177**



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Thank you very much!